Karthik Reddy
Data Scientist
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Professional Summary

Data Scientist with 3 years of experience delivering actionable insights and predictive models across healthcare and retail domains. Skilled in Python, SQL, machine learning, data visualization, and cloud-based analytics. Proven ability to translate complex data into business strategies that improve performance and customer engagement.

Technical Skills

Programming & Data Science: Python, R, SQL, C++, C, Java
ML/AI Tools & Libraries: Scikit-learn, TensorFlow, Pandas, Keras, NumPy
Visualization & Analytics: Tableau, Power BI, seaborn
Statistical Methods: Hypothetical Testing, ANOVA, Time Series, Confidence Intervals, Bayes Law, Principal Component Analysis (PCA), Dimensionality Reduction, Cross-Validation, Auto-correlation
Machine Learning: Regression analysis, Decision Trees, Random Forest, Support Vector Machines, KNN and Ensemble Method, Naive Bayes, K-Means Clustering, ANN, CNN, LSTM, XGBoost, Sentiment Analysis, Natural Language Processing (NLP).
Cloud & DevOps: AWS Cloud (S3, EC2, SageMaker), Docker, GitHub Actions (CI/CD)
Frame works: Flask

Professional Experience

Centene Corporation, Saint Louis, MO **Oct 2024 – Present**
Data Scientist
Domain: Healthcare

Responsibilities:

- Built a predictive model using Random Forest and XGBoost to identify high-risk patients, improving prediction accuracy by 18%.
- Designed an end-to-end data pipeline (ETL) using Python and SQL for data ingestion from EMR systems.
- Created Power BI dashboards for hospital administrators to monitor readmission trends and patient outcomes.
- Deployed machine learning models on AWS SageMaker for scalable inference.
- Collaborated with data engineers and clinicians to refine risk factors and optimize model interpretability using SHAP.

Environment: Python, SQL, Power BI, ETL, EMR, Scikit-learn, XGBoost, Pandas, NumPy, Microsoft Power BI, SHAP, AWS SageMaker, AWS S3, EC2, PostgreSQL / SQL Server, Git, Jira

Key Achievements:

Reduced readmission rates by 12% through data-driven interventions.
Improved model inference time by 25% after optimizing feature engineering pipeline.

PNB, India **Aug 2022 – Sep 2023**
Jr. Data Scientist
Domain: Banking

Responsibilities:

- Performed customer segmentation using K-Means and hierarchical clustering based on purchase behaviour.
- Built ARIMA and Prophet models for monthly sales forecasting with 92% accuracy.
- Cleaned and processed large transactional datasets (5M+ records) using Pandas and SQL.
- Created Tableau dashboards visualizing customer lifetime value (CLV), retention rates, and purchase trends.
- Presented insights to marketing teams, driving a 15% increase in personalized campaign ROI.

Environment: Python, SQL, Tableau, Pandas, NumPy, Scikit-learn, Prophet, Statsmodels, Apache Airflow, MySQL / SQL Server, Git

Key Achievements:

Reduced campaign cost per conversion by 10%.
Automated ETL workflows using Airflow, cutting report generation time from hours to minutes.

Projects:

- Detecting Spam Emails Using TensorFlow (NLP): Built a deep-learning model using TensorFlow with text preprocessing (tokenization, embedding, padding) to classify emails as spam or non-spam with high accuracy.
- Credit Card Fraud Detection (ML): Processed 280K+ transactions and built Random Forest models for fraud detection; handled class imbalance and evaluated using precision, recall, F1-score, MCC, and confusion matrix.
- House Price Prediction (ML): Built regression models with feature engineering and Scikit-learn pipelines on Kaggle dataset to improve prediction accuracy.

Certifications:

- Python for Everybody (University of Michigan – 5 courses)
- Certified Pega System Architect 8.7

EDUCATIONAL DETAILS:

Master's: New Jersey Institute of Technology – Data Science (2023-2025)
Bachelor's degree: KL University - : Electronics and communication Engineering (2019-2023)